

Math 8 — End of Q4 Test · ANSWER KEY

End-of-Q4 Content · 20 questions

Teacher key. Student answers may use equivalent formats (fractions, decimals, with or without units).

1. $v = \text{flambda}$. $f = 10 \text{ Hz}$, $\text{lambda} = 2 \text{ m}$. Speed?

Answer: 20 m/s

2. $v = 340 \text{ m/s}$, $f = 170 \text{ Hz}$. lambda ?

Answer: 2 m

3. $v = 3 \times 10^8 \text{ m/s}$, $\text{lambda} = 500 \text{ nm}$. Frequency? SN.

Answer: 6×10^{14} or 6e14

4. $\text{lambda} = 5 \text{ m}$, $v = 25 \text{ m/s}$. f ?

Answer: 5 Hz

5. Light $3 \times 10^8 \text{ m/s}$. Time to travel $1.5 \times 10^{11} \text{ m}$?

Answer: 500 s

6. Angle of incidence 30° . Reflection?

Answer: 30°

7. Convex lens: object beyond $2f$. Image is ____?

Answer: inverted or Inverted or i

8. $T^2 = r^3$. If T doubles, r ratio?

Answer: 1.587 or 1.59

9. $r = 4 \text{ AU}$. T in years?

Answer: 8 yr

10. $r = 9 \text{ AU}$. T in years?

Answer: 27 yr

11. Cylinder $r = 3$, $h = 5$. Volume ($\times \pi$).

Answer: 45pi or 45pi

12. Cone $r = 3$, $h = 4$. Volume.

Answer: 12pi or 12pi

13. Sphere $r = 3$. Volume.

Answer: 36pi or 36pi

14. Cylinder $r = 2$, $h = 10$.

Answer: 40π or 40π

15. Sphere $r = 6$.

Answer: 288π or 288π

16. Kai rides 12 mph for 4 hrs. Distance?

Answer: 48 mi

17. 15 m ladder, base 9 m. Height?

Answer: 12 m

18. Pool $10 \times 5 \times 2$ m. Volume?

Answer: 100 m^3

19. 12 V, 3 A. P?

Answer: 36 W

20. $\$15/\text{hr} \times 40$ hrs. Pay?

Answer: 600 or \$600 \$